

1. Introduction

In April 2026, I started the module “Machine Learning.” At the beginning of the course, I had only a limited understanding of machine learning and its applications. Although I was aware of its increasing use across many sectors, such as healthcare, finance, and customer care, my knowledge was theoretical. I had never had the opportunity to learn how machine learning models are designed, trained, and evaluated in practice. During the module, I had the opportunity to delve into several aspects of machine learning, such as traditional machine learning algorithms, clustering, artificial neural networks, convolutional neural networks, model selection, and evaluation.

In addition, I had the opportunity to work in a team to write an analytical report in which we attempted to apply regression models for pricing prediction. Through exercises, seminar activities, and individual presentations, I had the opportunity to start developing my skills on this topic.

In this reflection paper, I decided to use the structure proposed by Rolfe et al. (2001) to investigate my experience. In particular, I will reflect on the knowledge and skills I developed throughout the module, my experience of working as part of a team for the first assignment, and my final assignment, including both the benefits and challenges, and how these experiences contributed to my overall personal and professional development.

2. What I did during the module

The module introduced me to the fundamental technical knowledge of machine learning, its implementation, selection, and evaluation, its uses, and the ethical issues associated with it. During the course, I delved into all these aspects through theoretical

knowledge and practical activities, such as Jupyter Notebooks, a team project, and an individual project.

During the team project, I had the opportunity to work with other colleagues to implement linear regression models that aimed to predict house prices and listing segmentations. In this assignment, I participated in weekly meetings in which we discussed the tasks that needed to be completed, set clear goals, and regularly monitored our progress to ensure that the project remained on track. In this particular project, my task was to perform data cleaning and exploratory data analysis (EDA), which consisted of an exploration of the dataset highlighting some interesting outputs and data cleaning to improve the quality of the dataset before implementing linear regression models and clustering models.

In my individual presentation project, I compared two different machine learning models: one traditional and one deep learning. I chose the MovieLens 1M dataset and implemented an SVD Matrix factorization and a Neural Collaborative Filtering to create a recommender system.

3. Reflection on My Learning Experience

As stated above, this was my first experience with technical knowledge of machine learning, and for that reason, I was intimidated by the exercises included in the module. Additionally, this was my second time working on a university team project, and the fact that I did not know much about machine learning made me doubt the contributions I could make to the group. Throughout the project, I had the opportunity to do some programming exercises that helped me better understand new and previously studied topics, such as K-means and clustering. As in previous

modules, K-means and clustering were concepts that I had difficulty understanding, and having the opportunity to work on an exercise based only on this topic helped me learn more about them.

In addition, there were also topics that I particularly enjoyed studying and delving into, such as Artificial Neural Networks, which helped me to understand how information is processed through interconnected layers and how models learn from data, and Convolutional Neural Networks (CNNs), which helped me understand how convolutional layers automatically extract features.

The team project helped me improve my skills by learning from others' knowledge. In particular, our coordinator, Ariel, was extremely effective in organizing the group's work and managing conflicts. His supportive approach created a positive working environment in which I immediately felt comfortable contributing to discussions and sharing my ideas without worrying about making mistakes. As a result, I approached the project with greater confidence and became more willing to express my opinion.

The final project was more challenging than the team project. Unlike the group assignment, every aspect of the project, from selecting the data and machine learning models to interpreting the evaluation results, had to be completed independently. The absence of teammates with whom I could discuss ideas or validate my decisions made it more difficult.

I chose to develop a recommender system because I have always been interested in how companies use recommendation algorithms. However, throughout the project, I experienced several moments of uncertainty, particularly regarding the implementation of the SVD and Neural Collaborative Filtering models and the interpretation of

evaluation metrics. At times, I felt frustrated and unsure whether I was progressing in the right direction or not. To overcome these difficulties, I consulted additional literature and materials to verify whether I was following the correct implementation steps.

Despite these challenges, working through the problems independently strengthened my problem-solving skills and increased my confidence in researching, testing, and justifying my decisions.

4. Professional Development and Future Goals

The module allowed me to gain valuable experience in machine learning; in particular, I developed a basic foundation for my future academic and professional development. It was particularly fascinating to discover the numerous applications of these tools, as well as all the challenges and limitations that come with them.

During this experience, I learned many things as a student and as a future professional. It is still difficult for me to make decisions independently, so I hope to build greater confidence in working alone without feeling frustrated if the results are different from what I expected.

In future projects, I believe I can approach machine learning projects more systematically, having a better understanding of data preparation, model selection, evaluation, and the ethical implications of AI systems.

In conclusion, I hope to continue improving both my technical and collaborative skills, confident that, after this experience, I can approach future challenges with greater resilience and commitment to my professional development.

Reference list

Rolfe, G., Freshwater, D. and Jasper, M. (2001). Critical reflection for nursing and the helping professions. A user's guide. *Nursing standard*, 16(20).